proof for spinning, charged holes, 285
Price's theorem (how the hair gets lost), 280–85, 282f, 283f, 350
implications of hairlessness: "black-hole uniqueness," 27, 286, 425
quantum hair, 284
Hawking radiation from a black hole:
Zel'dovich's discovery of, for a rotating hole, 429–35
Hawking's discovery of, in general, 435–36
Zel'dovich's acceptance of, 439
based on quantum field theory in curved
spacetime, 436–39
properties of, 435–36
relationship to superradiance, 433n
as evaporation of black-hole atmosphere, 443–45, 443f
see also Black-hole evaporation
H-bomb: *see* Nuclear weapons research
Hoop conjecture:
formulation of, 266–67, 267f
evidence for, 264–67
Horizon of black hole:
overview of, 26, 28–30
name coined, 255
circumference of, 28–29, 28n
spin of, 291–94
maximum spin rate, 51–52, 293–94
shape of, 28, 51, 293, 293f
entropy of, 423–26, 442, 445–46
surface gravity of, 436
temperature of, 427, 436, 442, 445, 446
freezing of things near, 217–18, 239, 244–49, 255–56, 291–92
makes quantum fields exotic, 491–92
membrane-paradigm description of, 405–11, 406f, 41 0f, 443, 445
Hawking's area-increase theorem for, 413, 413n, 416–17, 422–23, 427, 462
laws of evolution of, 427, 436, 442, 446
apparent horizon vs. absolute horizon, 414–17, 415b
teleological evolution of absolute horizon, 417, 418b
apparent horizon as harbinger of singularity, 462–63
see also Critical circumference
Hyperspace, 55, 55f, 128–30, 214, 214f, 289, 291, 456–58, 457f, 458f, 484–86, 487f, 496f, 500–501, 501f

Implosion of star to form black hole:
overview, 27
limited by white-dwarf and neutron-star
masses, 159–61, 161f, 175–78, 177f
prevented by mass ejection, 196–97, 205–6, 210–11, 211f, 238
masses of parent stars, 205–6
Oppenheimer-Snyder predictions of, 211–19
Finkelstein's reference frame for, 245–46, 249, 255
simulations on a computer, 238–39, 240–41
parable of the ants describing, 246–49, 247f